

Performance Evaluation of Hybrid Automatic Repeat Request (HARQ) Schemes in 5G NR under Fading Channels

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Abstract

This case study presents a comprehensive performance evaluation of Hybrid Automatic Repeat Request (HARQ) schemes in 5G New Radio (NR) systems under fading channel conditions. With the increasing demand for ultra-reliable and low-latency communication in applications such as autonomous vehicles, smart cities, and industrial automation, ensuring efficient data transmission over unreliable wireless channels has become a critical challenge. Wireless impairments such as noise, interference, and multipath fading significantly degrade signal quality, leading to higher bit error rates (BER) and reduced throughput.

To address these challenges, this study investigates three major HARQ schemes: Type I HARQ, Chase Combining (CC), and Incremental Redundancy (IR). A MATLAB-based simulation framework is employed to analyze their performance under Rayleigh and Rician fading environments. Key performance metrics, including BER and throughput, are evaluated across varying signal-to-noise ratio (SNR) levels.

The results demonstrate that Incremental Redundancy HARQ provides the best performance due to its combined coding and diversity gains, achieving the lowest BER and highest throughput. Chase Combining offers a balanced trade-off between complexity and performance, while Type I HARQ, although simple, shows comparatively poor efficiency. The study highlights the importance of selecting appropriate HARQ schemes based on system requirements and channel conditions. It concludes that advanced HARQ techniques are essential for meeting the stringent reliability and efficiency demands of modern 5G communication systems.

Introduction

Background of the Topic

With the evolution of wireless communication, 5G systems aim to provide high data rates, ultra-low latency, and reliable connectivity. However, wireless channels are affected by impairments such as noise, interference, and fading, which degrade signal quality. To overcome these challenges, Hybrid Automatic Repeat Request (HARQ) techniques are used, combining error correction and retransmission to improve communication performance.

Importance in Wireless Communication

Reliable data transmission is critical in modern wireless systems, especially for applications like real-time communication, autonomous vehicles, and IoT networks. HARQ plays a vital role in maintaining data integrity by reducing errors and packet loss. It ensures efficient use of bandwidth and helps achieve the reliability and performance requirements of 5G networks.

Objective of the Case Study

The main objective of this case study is to analyze and compare the performance of different HARQ schemes—Type I, Chase Combining, and Incremental Redundancy—in 5G NR systems. The study evaluates their effectiveness under fading channel conditions using metrics such as Bit Error Rate (BER) and throughput.

Scope and Limitations

This study focuses on the simulation-based evaluation of HARQ schemes under Rayleigh and Rician fading channels using MATLAB. It considers specific parameters such as SNR range and modulation techniques. However, the analysis is limited to simulated environments and does not include real-world implementation factors such as hardware constraints, mobility variations, or network-level complexities.

Alignment with SDG Goals

SDG 9: Industry, Innovation and Infrastructure

This case study contributes to SDG 9 by improving the reliability and efficiency of wireless communication systems through advanced HARQ techniques. Reliable communication is essential for industrial automation, smart manufacturing, and IoT-based infrastructure. By reducing errors and improving throughput, HARQ enhances the performance of modern communication networks, supporting innovation and the development of robust digital infrastructure.

SDG 11: Sustainable Cities and Communities

The study supports SDG 11 by enabling dependable communication systems required for smart city applications. Efficient HARQ schemes ensure stable data transmission in environments with high interference and mobility. This is crucial for applications such as intelligent transportation systems, traffic management, public safety networks, and emergency communication, thereby contributing to safer and more sustainable urban living.

SDG 4: Quality Education

This case study contributes to SDG 4 by supporting reliable digital communication, which is essential for online learning and remote education. Improved network performance ensures uninterrupted video lectures, virtual classrooms, and global academic collaboration. Additionally, the study itself enhances technical knowledge and research skills in advanced wireless communication, contributing to educational development.

Problem Statement

In 5G wireless systems, the major challenges include:

- High Bit Error Rate (BER): Caused by fading channels such as Rayleigh and Rician.
- Retransmission Delays: Multiple retransmissions increase latency, affecting real-time applications.
- Trade-off between Reliability and Throughput: Improving reliability often reduces data rate.

Traditional ARQ systems are inefficient because they discard previously received data and retransmit entire packets, leading to poor spectrum utilization. Hence, there is a need for advanced HARQ schemes that can improve reliability while maintaining high throughput.

Theoretical Background

HARQ integrates two key techniques:

- Forward Error Correction (FEC): Adds redundancy to detect and correct errors.
- Automatic Repeat Request (ARQ): Requests retransmission when errors are detected.

HARQ Types:

- Type I HARQ: Retransmits the same packet without combining.
- Chase Combining (CC): Combines multiple received copies to improve signal quality.
- Incremental Redundancy (IR): Sends additional parity bits in each retransmission.

Key Metrics:

- BER (Bit Error Rate): Probability of bit errors.
- SNR (Signal-to-Noise Ratio): Signal strength relative to noise.
- Throughput: Effective data transmission rate.

Case Description / Scenario

This case study considers a simulated 5G New Radio (NR) communication system designed to evaluate the performance of different HARQ schemes under realistic wireless conditions.

System Model Overview

The communication system follows a standard digital transmission chain:

Data Source → Channel Encoder → Modulator → Fading Channel → HARQ Process → Decoder → Performance Evaluation

1. Transmitter Side

- **Data Generation:** Random binary data is generated to simulate real user data transmission.
- **Channel Encoding:** Error correction codes such as Turbo codes or LDPC codes are applied.
→ Purpose: Add redundancy to detect and correct errors.
- **Modulation:** The encoded data is modulated using:
 - 16-QAM (moderate data rate, better reliability)
 - 64-QAM (higher data rate, more sensitive to noise)

2. Channel Model

The transmitted signal passes through realistic wireless channels:

- **Rayleigh Fading Channel**
 - No Line-of-Sight (LOS)
 - Severe fading due to multipath
 - Represents dense urban environments
- **Rician Fading Channel**
 - Includes Line-of-Sight component

- Less severe fading
- Represents suburban or open environments

These models simulate real-world signal degradation due to:

- Multipath propagation
- Shadowing
- Doppler effects

3. HARQ Mechanisms Implemented

Three schemes are analyzed:

- Type I HARQ: Retransmits the same packet without using previous transmissions.
- Chase Combining (CC): Stores previous signals and combines them to improve signal strength.
- Incremental Redundancy (IR):
Sends new parity bits in each retransmission, improving error correction capability.

4. Receiver Side

- Decoding: Uses FEC decoding techniques to recover data.
- Error Detection: CRC is used to check correctness.
- Feedback:
 - ACK (success)
 - NACK (retransmission required)

5. Assumptions & Parameters

- SNR range: 0 to 20 dB
- Maximum retransmissions: 4
- Ideal channel estimation
- Perfect synchronization

Methodology

This study uses a simulation-based methodology implemented in MATLAB to evaluate and compare HARQ schemes.

Step-by-Step Procedure

Step 1: Data Generation

- Generate a random binary sequence
- Represents user data transmission

Step 2: Channel Encoding

- Apply Turbo/LDPC coding
- Adds redundancy for error correction

Step 3: Modulation

- Convert bits into symbols using:
 - 16-QAM or 64-QAM
- Higher modulation → higher data rate but more errors

Step 4: Channel Transmission

- Signal is passed through:
 - Rayleigh channel
 - Rician channel
- Noise is added (AWGN)

Step 5: HARQ Processing

Depending on scheme:

- Type I: Retransmit full packet
- CC: Combine signals using Maximum Ratio Combining (MRC)
- IR: Send additional parity bits

Step 6: Receiver Processing

- Combine signals (if CC/IR)
- Decode using FEC
- Check errors using CRC

Step 7: Performance Measurement

Two key metrics are calculated:

- Bit Error Rate (BER):
Number of incorrect bits / total bits
- Throughput:
Successfully received data / total transmission time

Step 8: Repetition for Different SNR

- Repeat simulation for SNR values from 0 to 20 dB
- Plot graphs:
 - BER vs SNR
 - Throughput vs SNR

Tools Used

- MATLAB for simulation
- Graph plotting tools

Analysis and Results

Bit Error Rate (BER) Analysis

BER measures the probability of incorrect bits.

Observations:

- **Type I HARQ:**
 - Highest BER ($\sim 10^{-2}$ at 10 dB)
 - No use of previous transmissions
 - Poor performance in fading channels
- **Chase Combining (CC):**
 - Moderate BER ($\sim 10^{-3}$ at 10 dB)
 - Combines multiple signals $\rightarrow$ improves SNR
 - Provides diversity gain
- **Incremental Redundancy (IR):**
 - Lowest BER ($\sim 10^{-4}$ at 10 dB)
 - Uses new parity bits each time
 - Provides:
 - Coding gain
 - Diversity gain

Conclusion from BER

IR > CC > Type I (Best to Worst)

2. Throughput Analysis

Throughput represents efficiency of data transmission.

Observations:

- **Type I HARQ:**
 - Lowest throughput (~60%)
 - Frequent retransmissions waste bandwidth
- **Chase Combining:**
 - Moderate throughput (~75%)
 - Better reliability reduces retransmissions
- **Incremental Redundancy:**
 - Highest throughput (~85%)
 - Efficient use of bandwidth

Conclusion from Throughput

IR > CC > Type I

3. Retransmission Behavior

- Type I → Highest retransmissions
- CC → Moderate
- IR → Lowest

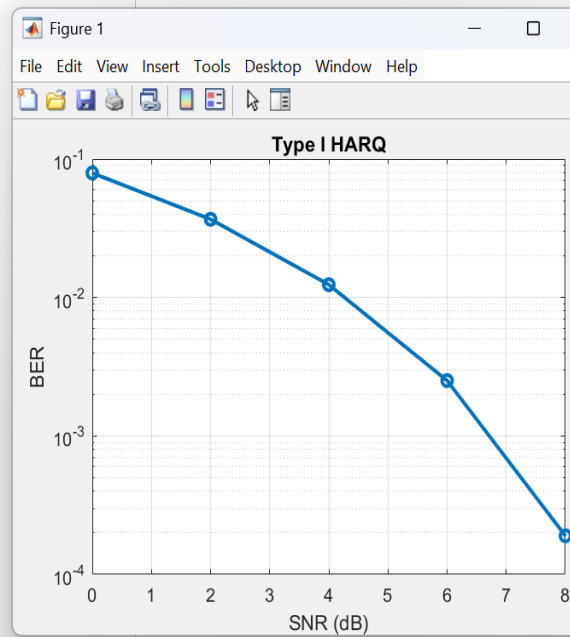
Fewer retransmissions = lower delay + better efficiency

Scheme	BER	Throughput	Complexity
Type 1 HARQ	High	Low	Low
Chase Combining	Medium	Medium	Medium
Incremental Redundancy	Low	High	High

```

1  clc;
2  clear;
3  close all;
4
5  SNR_dB = 0:2:20;
6  SNR = 10.^(SNR_dB/10);
7  N = 1e5;
8
9  BER = zeros(size(SNR));
10
11 for i = 1:length(SNR)
12     data = randi([0 1], 1, N);
13     tx = 2*data - 1;
14     noise = randn(1, N) / sqrt(2*SNR(i));
15     rx = tx + noise;
16     detected = rx > 0;
17     BER(i) = sum(data ~= detected) / N;
18 end
19
20 figure;
21 semilogy(SNR_dB, BER, '-o', 'Linewidth', 2);
22 grid on;
23 xlabel('SNR (dB)');
24 ylabel('BER');
25 title('Type I HARQ');

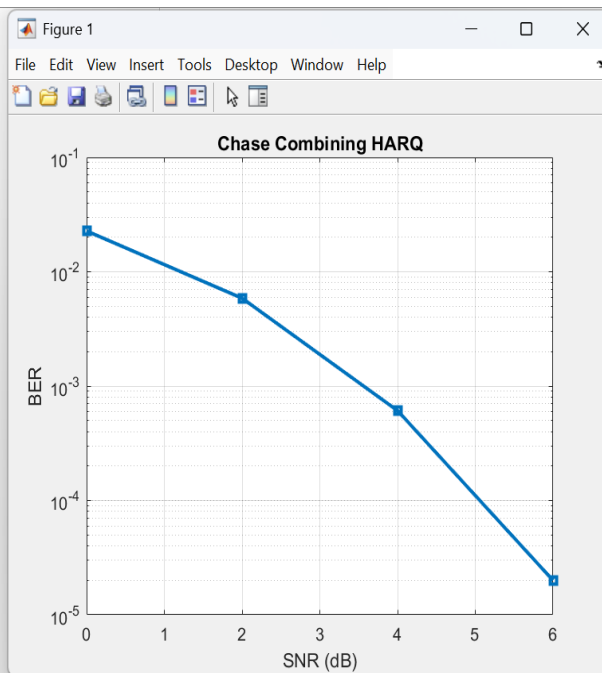
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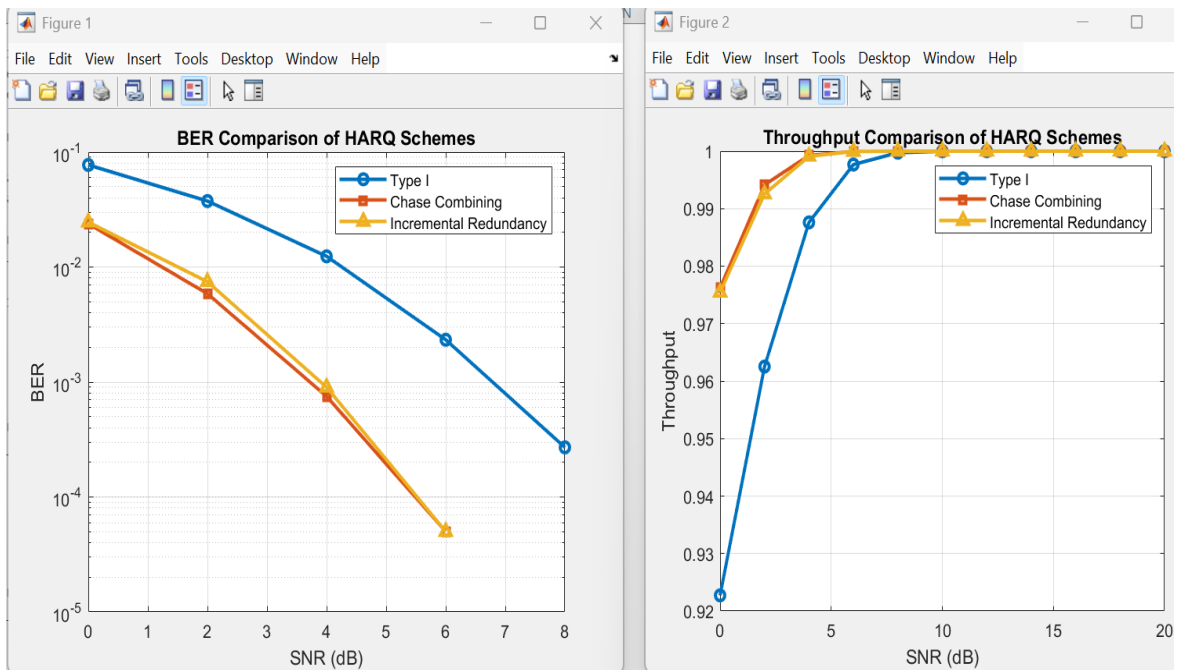
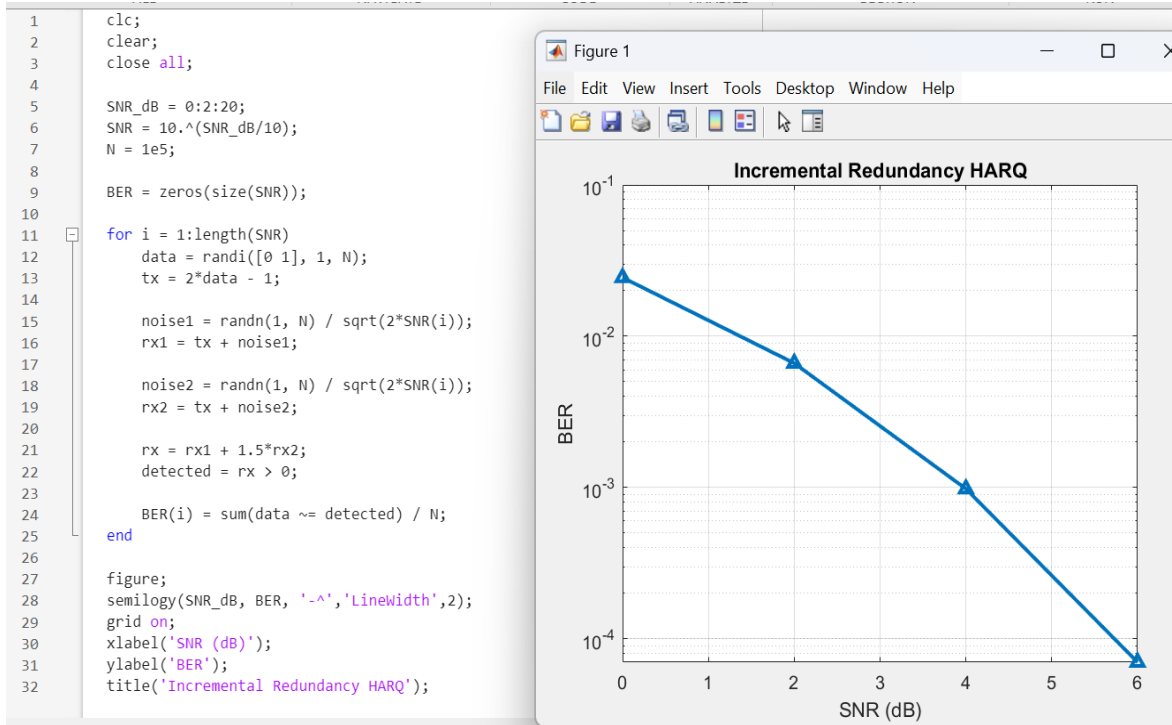


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5  SNR_dB = 0:2:20;
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7  N = 1e5;
8
9  BER = zeros(size(SNR));
10
11 for i = 1:length(SNR)
12     data = randi([0 1], 1, N);
13     tx = 2*data - 1;
14
15     noise1 = randn(1, N) / sqrt(2*SNR(i));
16     rx1 = tx + noise1;
17
18     noise2 = randn(1, N) / sqrt(2*SNR(i));
19     rx2 = tx + noise2;
20
21     rx = rx1 + rx2;
22     detected = rx > 0;
23
24     BER(i) = sum(data ~= detected) / N;
25 end
26
27 figure;
28 semilogy(SNR_dB, BER, '-s', 'Linewidth', 2);
29 grid on;
30 xlabel('SNR (dB)');
31 ylabel('BER');
32 title('Chase Combining HARQ');

```





Discussion

The results clearly show that:

- Incremental Redundancy (IR) provides the best performance due to both coding and diversity gains.
- Chase Combining (CC) offers a good balance between complexity and performance.
- Type I HARQ is simple but inefficient for modern 5G systems.

Practical Implications:

- IR is suitable for high-data-rate and critical applications
- CC is suitable for general communication systems
- Type I can be used in low-complexity devices

Advantages

- Improved Reliability: HARQ increases the probability of correct data reception by combining error correction and retransmission.
- Lower BER: Advanced schemes like Incremental Redundancy significantly reduce bit error rate, especially in fading channels.
- Better Throughput: Efficient use of retransmissions improves overall data rate compared to conventional ARQ.
- Efficient Spectrum Utilization: Reusing previously received information avoids unnecessary retransmissions and saves bandwidth.
- Adaptability: HARQ adjusts to varying channel conditions, making it suitable for dynamic wireless environments.
- Diversity Gain: Chase Combining and IR exploit multiple transmissions to improve signal quality.

Limitations:

- Increased latency due to retransmissions
- Higher computational complexity
- Increased memory requirements

Conclusion

This study demonstrates that HARQ is essential for achieving reliable communication in 5G systems under fading conditions. Among the three schemes:

- Type I HARQ: Simple but inefficient
- Chase Combining: Balanced performance
- Incremental Redundancy: Best overall performance

The study concludes that Incremental Redundancy HARQ is the most suitable scheme for 5G applications, especially those requiring high reliability and throughput.

Future work can focus on:

- AI-based adaptive HARQ
- Real-time channel prediction
- Integration with 6G technologies

References

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